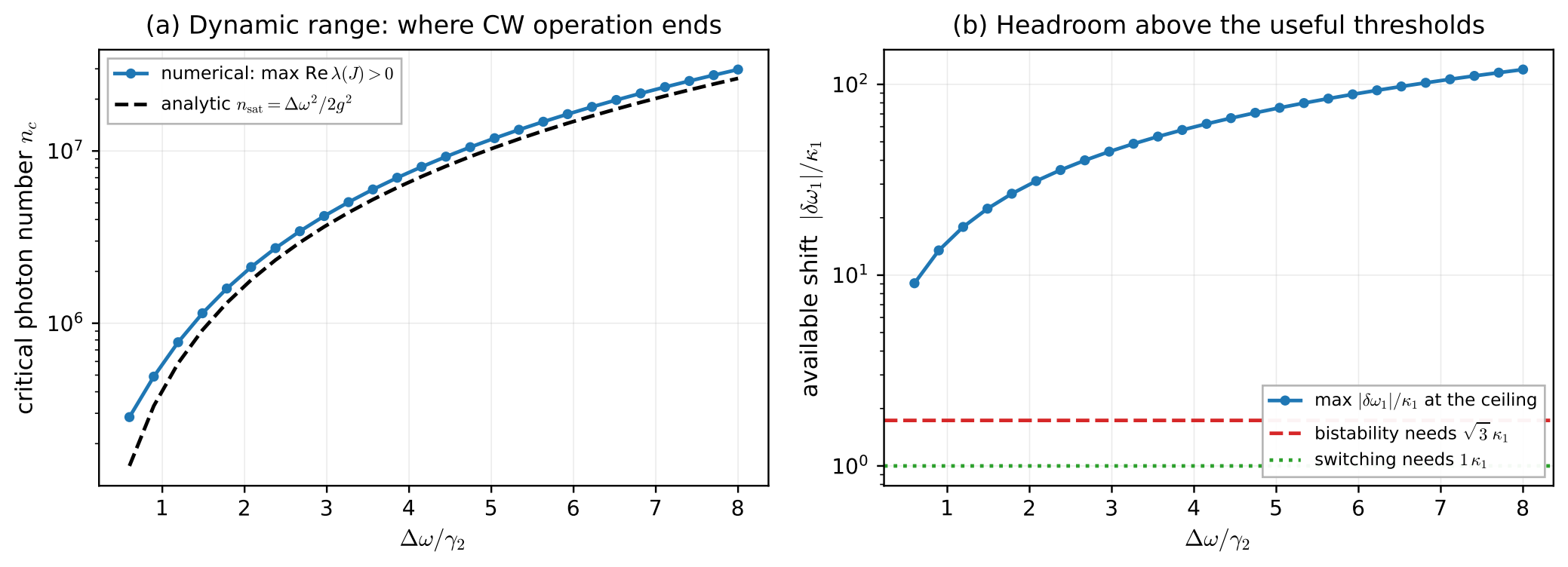
(a) Dynamic range: where CW operation ends
numerical: max Re $\lambda(J) > 0$
analytic $n_{\mathrm{sat}} = \Delta\omega^2/2g^2$
critical photon number n_c
10^7
10^6
1
2
3
4
5
6
7
8
$\Delta\omega/\gamma_2$
(b) Headroom above the useful thresholds
available shift $|\delta\omega_1|/\kappa_1$
10^2
10^1
10^0
max $|\delta\omega_1|/\kappa_1$ at the ceiling
bistability needs $\sqrt{3}\,\kappa_1$
switching needs $1\,\kappa_1$
1
2
3
4
5
6
7
8
$\Delta\omega/\gamma_2$